

Deepak Kumar Maurya

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Summary

Final-year **Computer Science & Engineering** student with hands-on experience in **Full-Stack Development, Scalable Backend Systems, Cloud-Native Applications, Database Management, Automation, and Machine Learning**. Proficient in **Python, C++, Java, SQL, JavaScript/TypeScript**, and modern full-stack frameworks including **React.js, Next.js, Node.js, and Express.js**, with expertise in developing **Production-Grade Web Applications** and **RESTful APIs**. Experienced with **Linux, AWS, Google Cloud Platform (GCP), Docker, GitHub Actions, and CI/CD** for designing, deploying, and maintaining scalable cloud-native solutions. Strong foundation in **DBMS, Distributed Systems, Software Engineering, Data Engineering, and DevOps**, complemented by multiple industry-recognized certifications in **Cloud Computing, Machine Learning, SQL, Data Engineering, GitHub, and Python Data Analytics**.

Education

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| Kalinga Institute of Industrial Technology
<i>Bachelor of Technology in Computer Science & Engineering</i> | July 2023 - May 2027
Bhubaneswar, Odisha |
| <ul style="list-style-type: none">• Relevant Coursework: Introduction to C Programming, Data Structures and Algorithms (C), Object Oriented Programming (Java), Operating Systems, Database Management System (Oracle Database Express Edition XE), Computer Network, Design and Analysis of Algorithms, Software Engineering• Relevant Elective: Linux System Administration (Red Hat Enterprise Linux), Computational Intelligence, High Performance Computing (MIPS), Cloud Computing, Deep Learning | |
| Royal Public Sr. Sec. School
<i>Intermediate in Non-Medical with Computer Science</i> | April 2021 - April 2022
Gurgaon, Haryana |
| <ul style="list-style-type: none">• Relevant Coursework: Introduction to Python Programming, Implementing Basic Data Structures in Python, Database Management System (MySQL), Connecting MySQL to Python | |

Experience

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| Technical Societies, Kalinga Institute of Industrial Technology
<i>Full-Stack Developer & DevOps Engineer (Google Developer Group, GeeksforGeeks & IoT Lab)</i> | Feb 2025 – Present
Bhubaneswar, Odisha |
| <ul style="list-style-type: none">• Developed and deployed 5+ production-grade full-stack applications using React.js, Next.js, Node.js, Express.js, Docker, AWS, and PostgreSQL, serving 2,000+ users.• Designed and implemented secure RESTful APIs, Role-Based Access Control (RBAC), and cloud-native deployment pipelines while collaborating with a team of 8+ developers.• Mentored 10+ students in Full-Stack Development, DevOps, Docker, and AWS, while contributing to hackathons, workshops, and university-wide technical initiatives. | |

Projects

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| GradeX LMS – AI-Powered Learning Management System <i>Next.js, TypeScript, Node.js, Drizzle ORM, PostgreSQL, AWS</i> |
| <ul style="list-style-type: none">• Designed and implemented a multi-tenant Learning Management System with secure Role-Based Access Control (RBAC) supporting 4 user roles, authentication, and session management.• Built an AI-powered assessment engine using the Gemini API for automated quiz generation and assignment evaluation.• Developed scalable cloud-native services integrating AWS SES, PostgreSQL, and Cloudflare R2 for automated notifications and secure file storage. |
| University Schedule Orchestration Platform <i>Next.js, Node.js, TypeScript, Express.js, PostgreSQL, Drizzle ORM, Docker, AWS</i> |
| <ul style="list-style-type: none">• Built a production-grade scheduling platform serving 2,000+ students with scalable RESTful APIs using Node.js, Express.js, and PostgreSQL.• Containerized and deployed the application on AWS EC2 using Docker and Nginx, achieving 100% uptime with automated health checks. |
| PrepCrate – Academic Resource Sharing Platform <i>Next.js, TypeScript, Node.js, PostgreSQL, Drizzle ORM, Cloudflare R2</i> |
| <ul style="list-style-type: none">• Developed a centralized academic resource platform enabling students to securely upload, organize, and access 500+ study materials.• Implemented a responsive Next.js frontend and integrated Cloudflare R2 with PostgreSQL via Drizzle ORM for scalable document management. |

Single Image Super-Resolution using CNN & GAN | Python, PyTorch, OpenCV

- Developed a deep learning pipeline using **CNNs** and **GANs** to reconstruct high-resolution images from low-resolution inputs.
- Implemented custom adversarial and perceptual loss functions to improve texture preservation and visual quality under academic guidance at KIIT.

Sentiment Analysis for Movie Reviews | Python, Scikit-learn, NLTK, Pandas

- Built a sentiment classification model achieving **89% accuracy** on **50,000 IMDB reviews** using TF-IDF feature engineering and NLP techniques.
- Evaluated **Logistic Regression** and **Naive Bayes** classifiers, optimizing hyperparameters to maximize model performance and F1-score.

Distributed File Storage System | Node.js, TypeScript, Express.js, PostgreSQL, Redis, Docker, AWS EC2

- Architected a distributed file storage platform with chunk-based replication, metadata indexing, and fault-tolerant recovery across multiple storage nodes.
- Implemented consistent hashing, heartbeat monitoring, automatic replica synchronization, and SHA-256 checksum validation to ensure high availability and data integrity.
- Deployed containerized services on AWS EC2 using Docker Compose and optimized file retrieval with Redis caching, reducing metadata lookup latency by approximately 60%.

Technical Skills

- **Programming Languages:** Python, C++, Java, C, SQL, TypeScript, JavaScript, HTML5, CSS3, SCSS
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Tailwind CSS, Scikit-learn, PyTorch, Pandas, NumPy
- **Databases & ORMs:** PostgreSQL, MySQL, MongoDB, Oracle SQL, Firebase Firestore, Supabase, Neon, Drizzle ORM, Prisma ORM, Mongoose
- **Cloud, DevOps & Infrastructure:** AWS (EC2, S3, SES), Google Cloud Platform (Cloud Run, App Engine, Compute Engine, Cloud Storage, Cloud Functions), Oracle Cloud Infrastructure (OCI), Docker, Docker Compose, Nginx, Cloudflare R2, GitHub Actions, CI/CD, Linux, Bash/Shell Scripting, Networking
- **Authentication & Security:** OAuth 2.0, JWT, Auth.js, Clerk, Firebase Authentication, Role-Based Access Control (RBAC)
- **AI, Machine Learning & Data Engineering:** Scikit-learn, TensorFlow, PyTorch, Linear Regression, Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naive Bayes, Random Forest, CNN, GAN, NLP, TF-IDF, Pandas, NumPy, ETL Pipelines, Apache Airflow, Data Warehousing, Data Modeling
- **Developer Tools & Platforms:** Git, GitHub, GitHub Actions, Postman, VS Code, Neovim, Arduino IDE, ESP32, STM32
- **Software Engineering Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Distributed Systems, RESTful APIs, API Development, Microservices, System Design, MVC Architecture, WebSockets, Socket Programming, Database Normalization, Reverse Proxy, Load Balancing, Concurrency, Multithreading, Agile Methodologies, Software Development Life Cycle (SDLC), Design Patterns, Automation, Data Analytics, Data Science, Big Data

Certifications

- **Cloud Computing**
 - Google Cloud Engineer Professional Certificate *Google Cloud – Coursera*
 - Developing Applications with Google Cloud *Google Cloud – Coursera*
 - Architecting with Google Compute Engine *Google Cloud – Coursera*
 - OCI Foundations Associate *Oracle Cloud Infrastructure*
 - OCI AI Foundations Associate *Oracle Cloud Infrastructure*
- **Data Engineering & Analytics**
 - Associate Data Engineer in SQL *DataCamp*
 - SQL Associate *DataCamp*
 - Python Data Associate *DataCamp*
 - Google Data Analytics with Python *Google – Coursera*
- **Software Engineering**
 - GitHub Foundations *GitHub*
 - Red Hat System Administration I (RH124) *Red Hat*
- **Artificial Intelligence**
 - Machine Learning Specialization *DeepLearning.AI & Stanford Online – Coursera*
 - Machine Learning Foundations: A Case Study Approach *University of Washington – Coursera*